

1A1 6.2

$$f \text{ na } V = \text{span} \left\{ \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} \right\} \subseteq \mathbb{R}^4$$

$$f((2 \ 1 \ 0 \ 1)^T) = (4 \ 1 \ 2 \ 5)^T$$

$$f((1 \ 0 \ 1 \ 2)^T) = (2 \ 2 \ -2 \ -2)^T$$

$$f^n((5 \ 1 \ 3 \ 7)^T) = ?$$

nalezi báze V je $B = \left(\begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} \right) = (v_1, v_2)$

$$\text{pak } [f((2 \ 1 \ 0 \ 1)^T)]_B = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$\sim [f((1 \ 0 \ 1 \ 2)^T)]_B = \begin{pmatrix} 2 \\ -2 \end{pmatrix}$$

$$\Rightarrow [f]_B = \begin{pmatrix} 1 & 2 \\ 2 & -2 \end{pmatrix}$$

nalezneme vlastní čísla:

$$\begin{pmatrix} 1-\lambda & 2 \\ 2 & -2-\lambda \end{pmatrix}; \det = (1-\lambda)(-2-\lambda) - 4 \Rightarrow \lambda = 2$$
$$\lambda = -3$$

tedy vlastní vektory

$$\lambda = 2: \begin{pmatrix} 1-2 & 2 \\ 2 & -2-2 \end{pmatrix} = \begin{pmatrix} -1 & 2 \\ 2 & -4 \end{pmatrix} \sim \begin{pmatrix} -1 & 2 \\ 0 & 0 \end{pmatrix} \Rightarrow M_2 = \text{span} \left\{ \begin{pmatrix} 2 \\ 1 \end{pmatrix} \right\}$$

$$\lambda = -3: \begin{pmatrix} 1+3 & 2 \\ 2 & -2+3 \end{pmatrix} = \begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix} \sim \begin{pmatrix} 0 & 0 \\ 2 & 1 \end{pmatrix} \Rightarrow M_{-3} = \text{span} \left\{ \begin{pmatrix} 1 \\ -2 \end{pmatrix} \right\}$$

tedy můžeme diagonalizovat $[f]_B$

$$[f]_B = P D P^{-1} = \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} 2^n & 0 \\ 0 & (-3)^n \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix}^{-1}$$

pozorování

(matice P je tvořena z vlastních vektorů)

~~$$(2 \text{ } 3 \text{ } 3) = \begin{pmatrix} 2^{n+1} \\ 2^{n+1} \end{pmatrix}$$~~

$$([f]_B)^n = \frac{1}{5} \begin{pmatrix} 2^{n+2} + (-3)^n & 2^{n+1} - 2(-3)^n \\ 2^{n+1} - 2(-3)^n & 2^n + 4(-3)^n \end{pmatrix}$$

potam $[(5 \ 1 \ 3 \ 7)^T]_B = [1 \ 3]^T$

take $[f^n((5 \ 1 \ 3 \ 7)^T)]_B = ([f]_B)^n \begin{pmatrix} 1 \\ 3 \end{pmatrix}$

$$= \frac{1}{5} \begin{pmatrix} 2^{n+2} + (-3)^n + 3 \cdot 2^{n+1} - 6(-3)^n \\ 2^{n+1} - 2(-3)^n + 3 \cdot 2^n + 12(-3)^n \end{pmatrix} = \begin{pmatrix} 2^{n+1} - (-3)^n \\ 2^n + 2(-3)^n \end{pmatrix}$$

$$\Rightarrow f^n((5 \ 1 \ 3 \ 7)^T) = (2^{n+1} - (-3)^n) \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} + (2^n + 2(-3)^n) \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix}$$

~~$$\begin{pmatrix} 5 \cdot 2^{n+1} \\ 2 \cdot 2^n - (-3)^n \\ 2^n + 2(-3)^n \\ 4 \cdot 2^{n+1} + 3 \cdot (-3)^n \end{pmatrix}$$~~

$$\Rightarrow \begin{pmatrix} 5 \cdot 2^{n+1} \\ 2 \cdot 2^n - (-3)^n \\ 2^n + 2(-3)^n \\ 4 \cdot 2^{n+1} + 3 \cdot (-3)^n \end{pmatrix} = f^n((5 \ 1 \ 3 \ 7)^T)$$